ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a number is divisible by 10	The part after 'if' is the hypothesis.
2	**(b)** its ones digit is 0	The part after 'then' is the conclusion.
3	**(c)** false	Counterexample: 2. 2 is prime and even.
4	**(d)** If a figure is a square, then it is a rectangle.	Subject becomes the hypothesis.
5	**(a)** 3	List the factors of 9 in pairs; the count is 3 (odd, so 9 is a perfect square).
6	**(c)** If a number is a multiple of 4, then it is a multiple of 8.	Interchange the hypothesis and the conclusion.
7	**(c)** 729	Digit sum of 729 is 18, a multiple of 9.
8	**(a)** True	The converse holds.
9	**(d)** two lines are parallel	The part after 'if' is the hypothesis.
10	**(a)** it is a rectangle	The part after 'then' is the conclusion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converse: If the ones digit of a number is 0, then the number is divisible by 10. (1); it is true (1).	1+1
12	If a figure is a square, then it has four equal sides (1); hypothesis: a figure is a square (1).	1+1
13	Factors: 1, 3, 9 (1); total 3; odd, so it is a perfect square (1).	1+1
14	Converse: If the alternate interior angles made by a transversal are equal, then the two lines are parallel. (1); it is true (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If the diagonals of a quadrilateral are perpendicular, then it is a rhombus. (1); Counterexample: a kite that is not a rhombus has perpendicular diagonals (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take $x = -4$ (1); $16 > 9$ but $-4 < 3$, so the statement is false (1).	1+1+1
17	$6^2 + 8^2 = 36 + 64 = 100$ (1); $10^2 = 100$ (1); Right-angled (1).	1+1+1
	OR: Converse: If a number is even, then it is a multiple of 4. (1); Counterexample: 6 is even but not a multiple of 4 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take 12 (1); 12 is divisible by 4 and 6 but not by 24, so the statement is false (1).	1+1+1
	OR: $5^2 + 12^2 = 25 + 144 = 169$ (1); $14^2 = 196$ (1); Not right-angled, as the two values differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$3 + 5 = 8$, which is even (1); Let the numbers be $2m + 1$ and $2n + 1$ (1); Sum $= 2m + 2n + 2 = 2(m + n + 1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $6 \times 7 = 42$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2
20	$7 + 11 = 18$, which is even (1); Let the numbers be $2m + 1$ and $2n + 1$ (1); Sum $= 2m + 2n + 2 = 2(m + n + 1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $9 \times 10 = 90$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** false	Counterexample: 9. $9 = 3 \times 3$.
2	**(c)** If a number is a multiple of 10, then it ends in 0.	Subject becomes the hypothesis.
3	**(c)** 6	List the factors of 12 in pairs; the count is 6 (even, so 12 is not a perfect square).
4	**(c)** If a number is a multiple of 5, then it is a multiple of 15.	Interchange the hypothesis and the conclusion.
5	**(d)** 4536	Digit sum of 4536 is 18, a multiple of 9.
6	**(b)** True	The converse holds.
7	**(b)** a parallelogram is a rectangle	The part after 'if' is the hypothesis.
8	**(d)** the alternate interior angles made by a transversal are equal	The part after 'then' is the conclusion.
9	**(c)** false	Counterexample: 6. 6 is even.
10	**(c)** If a triangle is equilateral, then it is isosceles.	Subject becomes the hypothesis.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If a number is a multiple of 5, then it ends in 0 or 5 (1); hypothesis: a number is a multiple of 5 (1).	1+1
12	Factors: 1, 3, 5, 15 (1); total 4; even, so it is not a perfect square (1).	1+1
13	Converse: If the square of the longest side of a triangle equals the sum of the squares of the other two sides, then the triangle is right-angled. (1); it is true (1).	1+1
14	If a quadrilateral is a rhombus, then it is a parallelogram (1); hypothesis: a quadrilateral is a rhombus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a number is divisible by 5, then it ends in 5. (1); Counterexample: 10 is divisible by 5 but does not end in 5 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take $n = 4$ (1); $2(4) + 1 = 9 = 3 \times 3$, so the statement is false (1).	1+1+1
17	$8^2 + 15^2 = 64 + 225 = 289$ (1); $17^2 = 289$ (1); Right-angled (1).	1+1+1
	OR: Converse: If a number is odd, then it is prime and greater than 2. (1); Counterexample: 9 is odd but not prime (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take $n = 11$ (1); $121 - 11 + 11 = 121 = 11^2$, so the statement is false (1).	1+1+1
	OR: $7^2 + 24^2 = 49 + 576 = 625$ (1); $26^2 = 676$ (1); Not right-angled, as the two values differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$9 + 13 = 22$, which is even (1); Let the numbers be $2m + 1$ and $2n + 1$ (1); Sum $= 2m + 2n + 2 = 2(m + n + 1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $13 \times 14 = 182$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2
20	$15 + 21 = 36$, which is even (1); Let the numbers be $2m + 1$ and $2n + 1$ (1); Sum $= 2m + 2n + 2 = 2(m + n + 1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $14 \times 15 = 210$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	List the factors of 16 in pairs; the count is 5 (odd, so 16 is a perfect square).
2	**(d)** If a number is divisible by 4, then it is divisible by 12.	Interchange the hypothesis and the conclusion.
3	**(a)** 810	Digit sum of 810 is 9, a multiple of 9.
4	**(c)** True	The converse holds.
5	**(b)** an integer n is even	The part after 'if' is the hypothesis.
6	**(b)** n^2 is odd	The part after 'then' is the conclusion.
7	**(a)** false	Counterexample: 2 and 3. $2+3=5$ is odd.
8	**(b)** If a number is a multiple of 6, then it is even.	Subject becomes the hypothesis.
9	**(c)** 6	List the factors of 20 in pairs; the count is 6 (even, so 20 is not a perfect square).
10	**(d)** If a number has an odd number of factors, then it is a perfect square.	Interchange the hypothesis and the conclusion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Factors: 1, 5, 25 (1); total 3; odd, so it is a perfect square (1).	1+1
12	Converse: If n^2 is even, then the integer n is even. (1); it is true (1).	1+1
13	If a number is even, then it is divisible by 2 (1); hypothesis: a number is even (1).	1+1
14	Factors: 1, 2, 3, 6, 9, 18 (1); total 6; even, so it is not a perfect square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If two angles are equal, then they are vertically opposite. (1); Counterexample: two angles of 60° in an equilateral triangle are equal but not vertically opposite (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take $\sqrt{2}$ and $-\sqrt{2}$ (1); $\sqrt{2}+(-\sqrt{2})=0$ is rational, so the statement is false (1).	1+1+1
17	$9^2+12^2=81+144=225$ (1); $16^2=256$ (1); Not right-angled, as the two values differ (1).	1+1+1
	OR: Converse: If the sum of the digits of a number is divisible by 9, then the number is divisible by 9. (1); The converse is also true (1); Both the proposition and its converse hold, so we may write the two as a single 'if and only if' statement (1).	1+1+1
18	A false statement needs only one counterexample (1); Take 11 and 3 (1); $11-3=8$, so the statement is false (1).	1+1+1
	OR: $20^2+21^2=400+441=841$ (1); $29^2=841$ (1); Right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$17+9=26$, which is even (1); Let the numbers be $2m+1$ and $2n+1$ (1); Sum $=2m+2n+2=2(m+n+1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $17\times 18=306$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2
20	$23+5=28$, which is even (1); Let the numbers be $2m+1$ and $2n+1$ (1); Sum $=2m+2n+2=2(m+n+1)$ (1); A multiple of 2, so even (2).	1+1+1+2
	OR: $22\times 23=506$, even (1); Of two consecutive integers one is even (1); So the product has the factor 2 (1); Hence even (2).	1+1+1+2

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